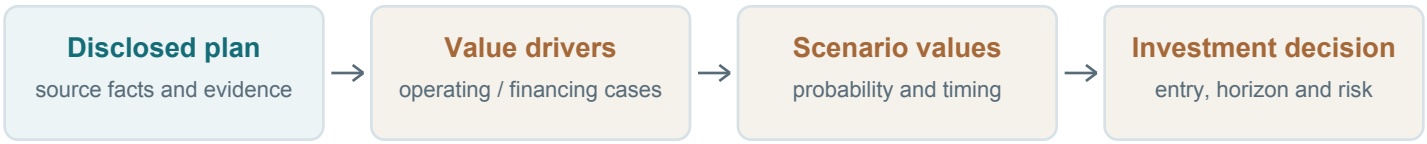
Modeling value creation after a 13D disclosure

Private Markets Modeling | Discussion 13 October 2026 | Evidence through 12 October

Investment question: can a disclosed investor's ownership, influence and value-creation plan support a credible upside/downside valuation and a disciplined public-equity investment decision?

The investable asset is a public stock. The connection to private-markets modeling is the treatment of governance, operating improvements, capital allocation, realization timing and exit value. ML can help prioritize opportunities; it must connect to an interpretable economic case.



Translate a stated plan into economic drivers

Disclosed mechanism	Modeling implication	Evidence still required
Operational improvement	Revenue growth, margins and reinvestment change future cash flows	Baseline financials, execution costs, milestones and achievable assumptions
Asset sale / separation	After-tax proceeds, stranded costs and remaining business value	Asset economics, transaction feasibility and capital deployment
Capital return / financing	Cash, debt, buybacks and share count alter equity value and risk	Capacity, covenants, funding terms and share-price assumptions
Board / governance change	Changes the probability and timing of strategy execution	Voting influence, agreements, company response and decision rights

Current progress and modeling boundary

The historical following baseline, a 667-event feature table and chronological input preparation are complete. No issuer-specific cash-flow valuation or calibrated scenario probabilities have been produced. The scenario framework below is the next proposed economic layer.

A disclosed stake does not establish control, and a stated objective does not establish implementation. Related parties may report the same shares. Manager reputation, HFR performance and prior filing activity are candidate context, not validated probabilities of successful value creation.

Evidence available and what it can support

A broad public-disclosure sample

Prepared evidence	What it supports	Limit
15,813 initial candidates, 2013-2025	Source-backed identifiers, ownership rows and prior disclosure histories	Factual staging; not a validated investment universe. [E2]
667 single-filing two-year events	Join company characteristics and filing facts to existing stock/proxy outcomes	Selective coverage from 2,107 scheduled events and 694 calculable labels. [E3-E4]
55 focused activist candidates; 556 other/unresolved	Compare mechanisms across a broader covered universe	Also 12 other Event-Driven and 44 other HFR candidates; identities remain provisional.
Dated size, industry and prior activity	Initial context for opportunity selection	Valuation fundamentals, liquidity and volatility inputs remain unfinished.

Why simple following is insufficient

Existing focused two-year evidence	Result	Meaning
56 calculable events	+8.66 pp mean excess	A favorable gross average with a wide uncertainty interval.
Typical event	-14.91 pp median excess	The positive mean does not describe the typical opportunity.
95% mean interval	-17.34 to +40.43 pp	Underperformance remains compatible with the evidence.

pp = percentage points. 2018-2025 exploratory research window; 504-session holding and modeled delayed opening entry. Equally weighted gross stock-minus-Russell-size-ETF-proxy returns, not account CAGR or risk-adjusted alpha. Horizons changed after earlier outcomes were inspected. This motivates selection and economic modeling, not proof of a repeatable edge. [E1]

Ownership and disclosure uncertainty

Ownership-percentage summaries are present for 557/667 events. The 110 missing cases include 14 unresolved security attributions. Three related reporters each disclosing 8% must not become a fabricated 24% position. The maximum reported percentage is a source summary, not verified aggregate economic influence. [E4]

LLM processing is proposed to interpret stated relationships and objectives with exact supporting passages. It cannot turn a plan into a realized outcome, infer unsupported control, or supply externally calibrated execution probabilities. Compustat financials and dated manager mappings still need assembly and validation.

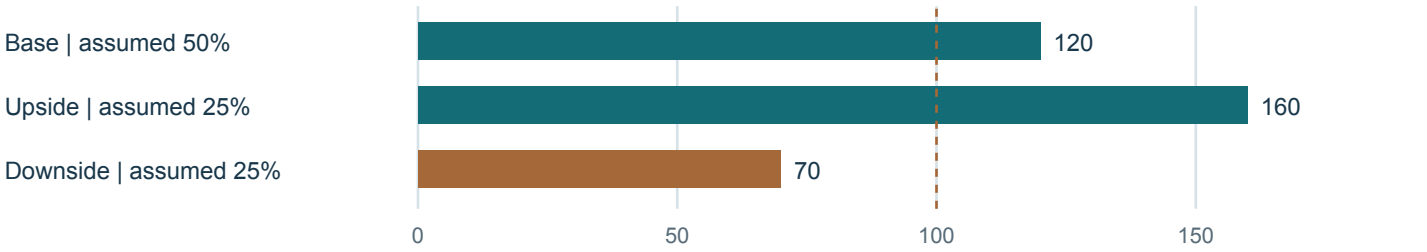
Proposed scenario and valuation model

For each selected case, establish a current operating baseline, specify the investor's requested changes, and model base, upside and downside paths. Separate value drivers, implementation probability and realization time. Use financial information available when the investment decision is made.

Model component	Proposed treatment
Operating cash flows	Revenue, margins, working capital, capital expenditure and restructuring costs under each path.
Enterprise to equity value	Discount operating value; reconcile debt, cash, taxes, contingent claims and dilution. Test terminal assumptions.
Execution and time	Evidence-based milestones and explicit uncertainty. Catalyst timing and probability remain assumptions to validate.
Investor return	Translate equity value and distributions into per-share investment outcomes under an explicit entry and exit convention.

Illustrative calculation only - not an issuer valuation

Assume an initial investment of 100, a common two-year horizon, and three terminal-wealth scenarios. The values and probabilities below are teaching assumptions; they are not learned, source-estimated or calibrated project results.



Illustrative terminal wealth; initial investment = 100

Expected terminal wealth = 0.50 x 120 + 0.25 x 160 + 0.25 x 70 = **117.50**. That is a 17.5% expected cumulative gain before costs. At an illustrative 10% annual required return, present value = 117.5 / 1.10^2 = **97.11**.

Expected future wealth above 100 can still be insufficient for the assumed required return. Probability, payoff and timing sensitivities therefore matter together. A forecast is useful only if the economic assumptions and downside are defensible.

Separate two kinds of uncertainty

Economic uncertainty: which operating/transaction path occurs, its payoff and timing. **Measurement uncertainty:** whether the extracted disclosure correctly identifies the investor, holdings and plan. Repeated LLM agreement measures stability; it does not establish the true probability of a value-creation outcome.

Acquisitions and delistings need source-backed settlement treatment. Replacement shares or contingent rights are not automatically immediate cash proceeds. Existing accounting exceptions limit some funded-portfolio results.

From economic forecasts to investment decisions

Decision	First comparison	Proposed modeling extension
Whether to invest	Relative-return selection under common rules	Require a credible valuation gap and downside/risk rationale.
When to enter	Existing delayed-entry convention	Compare disclosure reaction and later observable execution milestones.
How long to hold	Two-year target proposed	Relate holding period to catalyst timing and scenario realization.
How much to allocate	Existing 5% issuer targets and limits	Use loss exposure, uncertainty, liquidity and portfolio dependence.

These decisions jointly define a strategy; they do not automatically create one objectively optimal training label. A joint action model requires an explicit objective, costs and constraints. Initial work isolates selection before changing timing, horizon or sizing.

How ML and LLM processing support the model

A small feedforward neural network is planned to predict benchmark-relative return, compared with a training-mean reference, regularized regression and boosted trees. It may identify patterns across observable company and investor information. Its output is a statistical estimate, not a substitute for a cash-flow explanation.

Compare deterministic features, then validated LLM ownership/identity information, then disclosed intentions. Hold modeling and investment conventions comparable. Inspect extraction errors and repeated-run variability separately from predictive and economic performance.

Completed preparation and the next economic test

The 2023 development input table has 365 earlier outcome-mature training events, 86 later events and 69 encoded columns. All preprocessing uses earlier inputs. These examined historical data remain development evidence, not a fresh final test. [E5]

Next: select one source-supported mechanism; assemble dated fundamentals; build a transparent base/upside/downside case; then compare forecasts and investment decisions against the statistical selection model. Evaluate downside, holding time, realized benchmark-relative performance and implementation costs. No issuer-specific valuation or scenario calibration is complete yet.

Evidence and research status

Record	Supporting evidence
E1 | 24 September	Frozen historical following results and long-horizon trading conventions.
E2 | 24-25 September	Completed factual staging and audit: 15,813 initial-filing candidates.
E3-E5 | 12 October	Sample feasibility, feature/source joins and train-only preprocessing; exact local replay.

D1: Aswath Damodaran, Probabilistic Valuation: Scenario Analysis, Decision Trees and Simulations, NYU Stern. The numerical illustration uses hypothetical assumptions.

AI-assisted research tooling and writing. No predictive model, validated intent features or incremental investment result has been produced. Existing human-review records and historical outputs are preserved.